



MODEL

The stochastic multi-armed bandit

- A learner interacts with environment in **rounds**
- At each round, the learner chooses an **arm** to play, and receives a **reward** from the associated probability distribution
- Common assumptions:
 - There is a single optimal arm
 - The number of arms is small

We lift both assumptions

SETTING

We fix the time horizon T . At each round t , the learner **chooses an arm** a_t by either playing a past arm or picking a new arm from the reservoir $\mathcal{A}$. The learner gets a **reward** $Y_t \sim \nu_{a_t}$.

We aim for either **minimising cumulative regret**:

$$R(T) = \sum_{t=1}^T \mu^* - \mu_{a_t},$$

or for **identifying the best arm**, while minimising the probability of outputting a sub-optimal arm:

$$e(T) = \mathbb{P}(\hat{a}_T \notin \mathcal{A}^*).$$

BEST-ARM IDENTIFICATION

Algorithm: Elimination

- **Input:** $\bar{c}$
- **Initialise:** set $i \leftarrow 1$
- **while** $i < \log T / \bar{c}$ **do**
 Sample each arm in $\mathcal{A}_i$ a number of $t_i = \lfloor \bar{c}T / (|\mathcal{A}_i| \log T) \rfloor$ of times and compute their empirical means $(\hat{\mu}_i(a))_{a \in \mathcal{A}_i}$.
 Put in $\mathcal{A}_{i+1}$ the $1 \vee \lfloor |\mathcal{A}_i|/2 \rfloor$ arms that have highest empirical means and add on top of that $\lfloor |\mathcal{A}_i|/4 \rfloor$ new arms taken at random from $\mathcal{A}$.
 $i \leftarrow i + 1$
end
 Return any $\hat{a}_T \in \mathcal{A}_i$.

Upper bound

Theorem Set $\bar{c} = \log(4/3)$. Elimination satisfies

$$\mathbb{P}(\hat{a}_T \in \mathcal{A}^*) \geq 1 - 2 \log(T) \exp \left(-c \frac{\Delta^2 p^* T}{\log T} \right),$$

where $c = \bar{c}/19200$.

Lower bound

Theorem Consider $\Delta \in (0, 1/4)$ and $p^* \in [0, 1/4]$. For any bandit algorithm, there exists a bandit problem in $\mathfrak{B}_{\Delta, p^*}$ such that

$$e(T) \geq \frac{1}{4} \exp \left(-Tp^* \frac{\Delta^2}{32} \right).$$

NOTATION

- Potentially infinite set $\mathcal{A}$ called the **reservoir**
- Each arm $a \in \mathcal{A}$ is associated with a probability distribution ν_a supported on $[0, 1]$ with mean μ_a
- Highest mean $\mu^* = \max_{a \in \mathcal{A}} \mu_a$ and second highest mean $\mu_{sub} = \sup_{a \in \mathcal{A}: \mu_a \neq \mu^*} \mu_a$
- Minimal gap $\Delta = \mu^* - \mu_{sub}$; we assume $\Delta > 0$
- There exists a partition $\mathcal{A} = \mathcal{A}^* \cup \mathcal{A}_{sub}$
- Proportion p^* of optimal arms
- $\mathfrak{B}_{\Delta, p^*}$: set of bandit problems of which the proportion of optimal arms is at least p^* and the suboptimality gap is at least Δ

CUMULATIVE REGRET MINIMISATION

Algorithm: Sampling UCB

- **Input:** $\gamma \in (0, 1), L \geq 1$
- **Initialise:** Pick $\mathcal{L}$, with $|\mathcal{L}| = L$ arms from $\mathcal{A}$. Sample each arm once.
- **for** $t = L + 1$ **to** T **do**
 Compute for each arm $a \in \mathcal{L}$ the quantity

$$U_a^t = \hat{\mu}_a^t + \sqrt{\frac{\frac{\gamma^2}{4(1-\gamma)} + \log(\frac{\pi^2}{6}) + 2 \log(N_a^t)}{2N_a^t}}$$

Play $a_t = \arg \max_{a \in \mathcal{L}} U_a^t$
end

Upper bound

Theorem For $T \geq 2, \gamma \in (0, 1)$ and $L = \lceil 4 \log(T) / (p^* \gamma^2) \rceil$, the expected cumulative regret of Sampling UCB is upper bounded as:

$$\mathbb{E}R(T) \leq O \left(\frac{\log(T) \log(1/\Delta)}{p^* \Delta} \right).$$

Lower bound

Theorem Consider $\Delta \in (0, 1/4)$ and $p^* \in (0, 1/4]$. For any bandit algorithm, there exists a bandit problem in $\mathfrak{B}_{\Delta, p^*}$ such that

$$\mathbb{E}R(T) \geq \min \left(\frac{1}{60} \frac{\log(\Delta^2 T / 16)}{p^* \Delta}, \sqrt{T} \right).$$

ADAPTING TO p^*

In **regret minimisation** it is **impossible to adapt to p^*** , as follows from the following theorem.

Theorem Let $p^* \leq 1/4$ and $c > 0$ such that

$$T \geq 4 \left(\frac{c \log(T)}{p^* \Delta^2} \right)^2.$$

For any bandit algorithm $\mathfrak{A}$ such that for all bandit problems $\mathfrak{B}_{\Delta, p^*}$, we have

$$\mathbb{E}R(T) \leq \frac{c \log(T)}{p^* \Delta},$$

one has that $\forall q^* \leq \frac{4p^*}{c}$ there exists a problem in $\mathfrak{B}_{\Delta, q^*}$ such that

$$\mathbb{E}R(T) \geq \frac{\sqrt{T} \Delta}{4}.$$

In **best-arm identification**, adapting to p^* is possible, as is done in our elimination algorithm.

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